

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Computer Science and Engineering

CSE 4102: Computer Graphics and Image Processing Laboratory
4th Year 1st Term

SENTINEL

A Fully Procedural Game Engine and Open-World Military Shooter

Submitted by

Md. Shifat Hasan

Roll: 2107067

Submitted to

Md Tajmilur Rahman

Md. Mubtashim Abrar Nihal

Instructors, Department of Computer Science and Engineering



Department of Computer Science and Engineering
Khulna University of Engineering & Technology
Khulna 9203, Bangladesh

October 7, 2026

Abstract

Most video games ship with gigabytes of files made by artists: 3D models, photographs for surfaces, recorded sounds. *Sentinel* takes the opposite route. It is a complete game engine and an open-world military shooter in which *nothing* is loaded from a model, image or sound file. Every tree, building, soldier, vehicle, surface, cloud and sound effect is created while the program starts, from short mathematical recipes and a random seed. Only the small text files holding the shader programs are read from disk.

The engine is written in the Odin language on OpenGL 4.1, the newest version macOS offers (it has no compute shaders), and is organised in six layers that may only depend on the layers beneath them. A deferred renderer records what every pixel is made of and then lights it with a sun and moon, spot and point lamps, cascaded shadow maps, ambient occlusion, screen-space reflections, an atmospheric sky with wind-driven clouds, fog, bloom, tone mapping, colour grading and anti-aliasing. Surfaces come from 18 texture recipes built on periodic gradient noise, domain warping and cellular noise; normal maps are derived from height fields with a Scharr filter. Objects and characters are tables of primitives with noise deformers.

On top of the engine sit collision, a ladder state machine, four ground vehicles and a helicopter, enemy soldiers with stealth rules, synthesised audio, and a five-objective rescue mission in the spirit of *Project I.G.I.* The work was measured and verified, not only drawn: at 1080p the GPU needs 10.9 ms per frame, and 281 unit tests, 30 GL checks, 8 318 texture checks (including exact tiling of every material), 318 567 renderer checks and two scripted full playthroughs pass. The report explains every technique in plain language, with its governing equations, diagrams and screenshots, and documents a debugging case in which a star in the fog colour painted a bright vertical line across the night scene.

Keywords: procedural generation, game engine, OpenGL, deferred rendering, physically based shading, noise, texture synthesis, real-time graphics.

Contents

	PAGE
Abstract	i
Contents	ii
List of Tables and Figures	iii
CHAPTER I Introduction	1
1.1 Introduction	1
1.2 Background and Problem Statement	1
1.3 Objectives	1
1.4 Scope	2
1.5 Unfamiliarity of the Problem and its Solution	2
1.6 Project Planning and Work Distribution	2
1.7 Applications of the Work	2
1.8 Organization of the Project	2
CHAPTER II Related Works	3
2.1 Introduction	3
2.2 Related Works	3
2.3 Discussion	3
CHAPTER III Methodology	4
3.1 Introduction	4
3.2 Detailed Methodology	4
3.3 Conclusion	7
CHAPTER IV Implementation, Results and Discussions	8
4.1 Introduction	8
4.2 Experimental Setup	8
4.3 Evaluation Metrics	8
4.4 Dataset	8
4.5 Implementation and Results	9
4.6 Objective Achieved	10
4.7 Conclusion	10
CHAPTER V Societal, Health, Environment, Safety, Ethical, Legal and Cultural Issues	11
5.1 Intellectual Property Considerations	11
5.2 Ethical Considerations	11
5.3 Safety Considerations	11
5.4 Legal Considerations	12
5.5 Impact on Societal, Health and Cultural Issues	12
5.6 Impact on the Environment and Sustainability	12
CHAPTER VI Addressing Complex Engineering Problems and Activities	13
6.1 Complex Engineering Problems	13
6.2 Complex Engineering Activities	13
CHAPTER VII Conclusions	14
7.1 Summary	14
7.2 Limitations	14
7.3 Recommendations and Future Works	14
Appendix A: Equation Guide	15
References	20

List of Tables

Table No.	Description	Page
1.1	Tools and technologies used	2
2.1	Comparison with related approaches (typical characteristics)	3
4.1	Experimental setup	8
4.2	GPU time per pass, 1080p benchmark	9
6.1	Complex engineering problem attributes	13

List of Figures

Figure No.	Description	Page
1.1	Work plan by day (the single author performed every task).	2
3.1	The six layers (dependencies point downward) and the main loop.	4
3.2	Noise ingredients (illustration of the shader equations): gradient, fractal, warped, ridged, cellular.	5
3.3	The passes of one frame, left to right.	6
3.4	A view ray meets each cloud plane; the noise slides with the wind.	7
4.1	GPU time per pass (total 10.91 ms of the 16.7 ms budget).	9
4.2	The engine and game in action; (h) and (i) show the streak bug and its fix.	10
6.1	The verification loop used for every feature.	13

CHAPTER I

Introduction

1.1 Introduction

Imagine a film studio that buys no props: a recipe book tells assistants how to build every tree and cloud, and a different “seed” gives a different but believable forest. *Sentinel* is that studio for a game: a *game engine* (the machinery that draws, simulates and sounds a world) plus a first-person military mission. *Procedural* means shapes, colours and sounds are calculated by code, not stored in files. (A **GPU** is a second processor with thousands of tiny cores; a **shader** is the small program each core runs per pixel.)

1.2 Background and Problem Statement

Graphics courses teach lighting, textures and noise one at a time; a product needs them working *together* 60 times a second, usually with art loaded from files. The problem: *can a complete, good-looking, playable 3D game be produced by code alone on a student laptop?* Limits: no pre-saved model, image or precomputed file, and OpenGL 4.1 without compute shaders.

1.3 Objectives

1. Build a layered engine with downward-only dependencies.
2. Generate all meshes, textures and sounds from code and seeds.
3. Implement a modern renderer (PBR, shadows, sky).
4. Build a playable game with vehicles, ladders, doors and a mission.
5. Verify it with tests and a 16.7 ms frame budget.

1.4 Scope

The scope is the whole engine and one game (Table 1.1); networking, multiplayer and hand-made art are out of scope.

Table 1.1: Tools and technologies used

Tool	Role (plain words)
Odin language	The project language: fast like C, readable like a script.
OpenGL 4.1 core	The standard way to talk to the graphics card on macOS.
GLFW, miniaudio	Window and input; sound output.

1.5 Unfamiliarity of the Problem and its Solution

The design follows published theory, not an existing code base: commercial engines import art, and demo-scene shooters are not readable. Here “no files” is enforced everywhere.

1.6 Project Planning and Work Distribution

One student did all the work; Fig. 1.1 is rebuilt from the 138 commits.

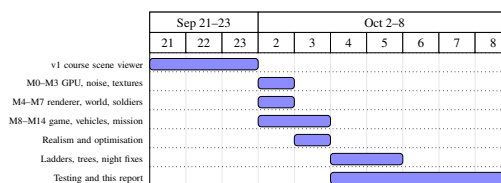


Figure 1.1: Work plan by day (the single author performed every task).

1.7 Applications of the Work

A teaching tool and a base for small games.

1.8 Organization of the Project

Chapters II to IV give related work, design and results; V and VI discuss social and engineering issues; VII concludes.

CHAPTER II

Related Works

2.1 Introduction

Sentinel combines noise, physically based shading, real-time pipelines and procedural games.

2.2 Related Works

Noise. Perlin’s noise [1] blends random gradients on a grid; summing scales gives natural detail. Worley’s cellular noise [2] gives pebbles and leaves; domain warping [13] bends patterns to hide the grid. Sentinel uses all three, made *periodic* so textures tile.

Shading. Cook–Torrance with GGX [3–5] is the standard reflection model; Oren–Nayar [8] suits rough cloth.

Pipeline and games. Deferred shading [14], cascaded shadows [7], Hillaire’s sky [6], ACES [9] and FXAA [11] form the pipeline; *.kkrieger* (2004) generated a shooter at run time and *Project I.G.I.* (2000) inspired the mission.

2.3 Discussion

Table 2.1 shows the gap.

Table 2.1: Comparison with related approaches (typical characteristics)

Feature	Tutorials	Commercial	.kkrieger	Sentinel
No imported art	Mixed	No	Yes	Yes
Complete game loop	No	Yes	Yes	Yes
Day/night, clouds, shadows	Rare	Yes	No	Yes
Works without compute shaders	Varies	Rarely	Yes	Yes
Written to be read	Yes	No	No	Yes

CHAPTER III

Methodology

3.1 Introduction

One frame is an assembly line: the game decides what exists, the engine builds it, the renderer paints it.

3.2 Detailed Methodology

3.2.1 Overall Framework

Figure 3.1: a lower layer never knows about a higher one. Source/Main.odin is only a table of contents: initialise, then repeat *input*, *update*, *render*, then shut down. Procedures stay near 40 lines and every GL call is checked.

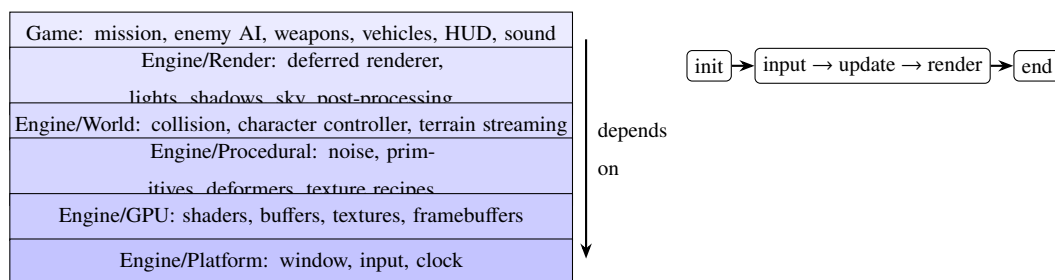


Figure 3.1: The six layers (dependencies point downward) and the main loop.

3.2.2 Procedural Geometry and Characters

Every object is a *table* of parts, each a primitive (box, sphere, cylinder, cone, capsule, torus, wedge), a transform, optional deformers and a material. A noise deformer moves each vertex

along its normal $\hat{n}$ by a smooth random amount of at most A , so a perfect sphere becomes a lumpy rock:

$$\mathbf{p}' = \mathbf{p} + A \nu(f\mathbf{p}) \hat{n}, \quad \|\mathbf{p}' - \mathbf{p}\| \leq A. \quad (3.1)$$

Soldiers are capsule skeletons; legs use two-bone IK: for upper and lower bone lengths a, b and hip-to-target distance c , the law of cosines gives the hip angle α between the target direction and the upper bone

$$\cos \alpha = \frac{a^2 + c^2 - b^2}{2ac}. \quad (3.2)$$

3.2.3 Procedural Textures

Each material is a *recipe*: a small shader that answers “what is the colour, bumpiness, shininess and crevice-darkness at this point?”. Each recipe is baked at start-up into a 1024×1024 picture (22 materials from 18 recipes: concrete, rock, sand, dirt, asphalt, metals, paints, wood, bark, leaves, needles, grass, canvas, camo, rubber, skin, glass) and keeps it on the GPU.

Gradient noise (a rubber sheet pinned at random slopes) blends the four corner gradients of a grid cell with the smooth curve $s(t) = 6t^5 - 15t^4 + 10t^3$, scaled by $\sqrt{2}$ so values lie in $[-1, 1]$. Fractal noise sums octaves, each at double frequency and half strength:

$$F(\mathbf{x}) = \frac{\sum_{i=0}^{n-1} 2^{-i} \nu(2^i \mathbf{x})}{\sum_{i=0}^{n-1} 2^{-i}}. \quad (3.3)$$

Choosing gradients from the cell index *modulo the period* P gives $F(\mathbf{x} + P\mathbf{e}) = F(\mathbf{x})$: the picture tiles with no seam (tested exactly). *Domain warping* $F(\mathbf{x} + s[F_1(\mathbf{x}), F_2(\mathbf{x})])$ swirls the pattern (stone, wood, soil); *ridged noise* $\sum a_i(1 - |v_i|)^2$ makes cracks and bark; *cellular noise* gives leaves and pebbles (Fig. 3.2).

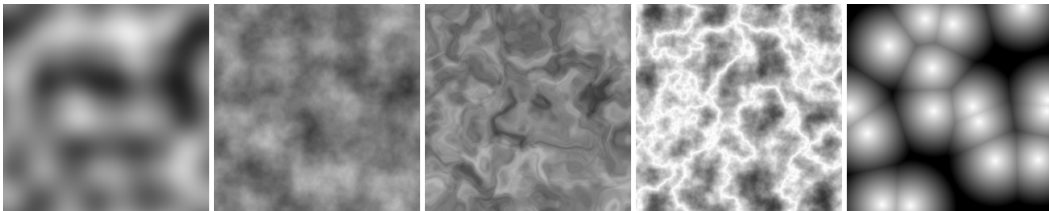


Figure 3.2: Noise ingredients (illustration of the shader equations): gradient, fractal, warped, ridged, cellular.

Bumps without geometry. A recipe’s height map becomes a *normal map* that fools the lighting into seeing bumps. The slope uses the Scharr 3×3 filter, the most direction-fair small filter, and crevices (height below the neighbours’ average) darken the albedo and ambient occlusion:

$$h_x \approx \frac{3(h_{NE} + h_{SE} - h_{NW} - h_{SW}) + 10(h_E - h_W)}{32 \Delta N}, \quad \hat{n} = \frac{(-k h_x, -k h_y, 1)}{\|\cdot\|}, \quad (3.4)$$

where Δ is one texel and N the cells per tile, so bump strength k is scale-independent. Triplanar mapping (weights $w_i = |n_i|^4 / \sum_j |n_j|^4$) avoids stretching.

3.2.4 The Rendering Pipeline

Deferred shading photographs a “data poster” of the scene first (each pixel’s colour, direction and shininess) and lights it once, which makes many lights cheap. The poster is the *G-buffer*: albedo, octahedral-packed normal with roughness and metalness, emission with occlusion, and depth (from which position is rebuilt). Fig. 3.3 lists the passes. Direct light uses the Cook–Torrance model [3]: a diffuse part plus a mirror-like specular part,

$$f = \frac{(1 - F_l)(1 - F_v) \mathbf{c}_d}{\pi} + D V F, \quad D = \frac{\alpha^2}{\pi((\mathbf{n} \cdot \mathbf{h})^2(\alpha^2 - 1) + 1)^2}, \quad F = F_0 + (1 - F_0)(1 - \mathbf{v} \cdot \mathbf{h})^5, \quad (3.5)$$

where $\mathbf{c}_d$ is the diffuse colour, D counts tiny mirror facets facing the viewer (GGX), $V = G / (4 \mathbf{n} \cdot \mathbf{l} \mathbf{n} \cdot \mathbf{v})$ is the Smith visibility (facets shading each other), F is glancing-angle reflection (Schlick, $F_0 = 0.04$ for non-metals), F_l, F_v are F at the light and view angles, and α is roughness squared. Distant things fade into haze, $C = T C_{\text{surf}} + (1 - T) C_{\text{haze}}$, $T = e^{-kd}$. Finally the HDR image is compressed to screen range by the ACES curve [9], graded, dithered [10] and smoothed (FXAA [11]):

$$C_{\text{out}} = \text{clamp}\left(\frac{x(2.51x + 0.03)}{x(2.43x + 0.59) + 0.14}\right), \quad x = \text{exposure} \times \text{radiance}. \quad (3.6)$$

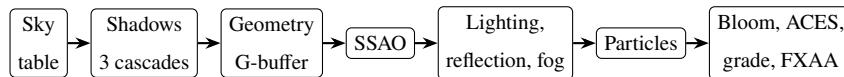


Figure 3.3: The passes of one frame, left to right.

3.2.5 Sky, Time of Day and Clouds

The sun's height follows astronomy, $\sin(\text{elev}) = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos h$ (latitude φ , declination δ , hour angle h). The sky colour comes from Rayleigh and Mie scattering [6]; the moon and a star field take over at night. Clouds live on two planes, cumulus at $H = 1500$ m and cirrus at 7 km. A view ray with direction $\mathbf{d}$ meets the plane at

$$\mathbf{P}_{xz} = \frac{H}{d_y} (d_x, d_z) + \mathbf{w} t, \quad (3.7)$$

where $\mathbf{w}$ is the wind velocity: shifting the noise by $\mathbf{w}t$ makes the clouds drift (Fig. 3.4).

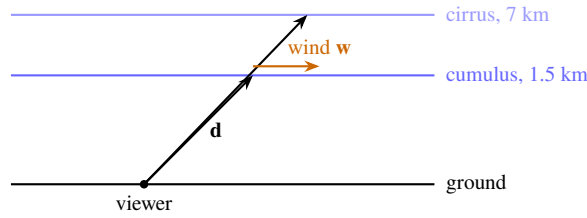


Figure 3.4: A view ray meets each cloud plane; the noise slides with the wind.

3.2.6 Terrain, Trees, Physics and Gameplay

Terrain is warped, ridged fractal noise flattened into a plateau for the base and streamed in 64 m chunks; grass, dirt, rock and sand blend by slope and height. **Trees** (oak, pine, birch, cypress) are tables of limbs and leaf lumps; each instance gets its own size, lean and tint, and the vertex shader bends it as a cantilever, $\Delta_x = 2s(0.04 + 0.1g)(y/9)^2$ (Δ_z is 35% of it), where $g = \frac{1}{2} + \frac{1}{2} \sin(1.3t + 0.05x + 0.03z)$ is a travelling gust.

Collision uses rotatable boxes and a capsule player. Cars follow the bicycle model, $\dot{\psi} = v \tan \delta / L$, and engine power fades toward top speed, $\dot{v} = a_0(1 - 0.6|v|/v_{\max})$. **Ladders** are a four-state machine (none, mounting, climbing, topping out) in which the controller owns the body, so it cannot stick. A staged machine drives five mission goals; sounds are synthesised.

3.3 Conclusion

Small data-driven techniques give a deterministic world.

CHAPTER IV

Implementation, Results and Discussions

4.1 Introduction

This chapter states how the work was measured and shows what it produces.

4.2 Experimental Setup

An Apple-silicon Mac (macOS, OpenGL 4.1 core), release build (`-o:speed`): 164 Odin files (about 19 200 lines) and 47 GLSL files (about 2 000 lines). Benchmark: a scripted 1 km flight at 1080p, 300 frames (Table 4.1).

Table 4.1: Experimental setup

Item	Value
Machine	Apple-silicon Mac, macOS, OpenGL 4.1 core profile
Build	Odin dev-2026-09, <code>-o:speed</code>
Code size	164 Odin files (about 19 200 lines), 47 GLSL files (about 2 000 lines)
Resolution	1920×1080 benchmark; 2560×1440 screenshots

4.3 Evaluation Metrics

Frame time $t = 1000/\text{fps}$ ms (budget 16.7 ms) and GPU time per pass; test pass or fail; and tiling error $\max |f(\mathbf{u} + \mathbf{k}) - f(\mathbf{u})|$, which must be zero.

4.4 Dataset

No dataset is used. All content comes from fixed seeds, so every run is reproducible and no image, model or recording is stored.

4.5 Implementation and Results

4.5.1 Quantitative results

The display locks the average frame to 16.75 ms (60 Hz); the GPU work needs only 10.91 ms (Table 4.2, Fig. 4.1). Verification: 281 unit tests (noise, meshes, IK, collision, ladders, vehicles, mission, audio), 30 GL checks, 8 318 texture checks (exact tiling of every material), 318 567 renderer checks (lighting, shadows, atmosphere, instancing) and two scripted playthroughs that both finish 5 of 5 objectives.

Table 4.2: GPU time per pass, 1080p benchmark

Pass	Time (ms)	Share
Geometry (G-buffer)	4.11	38%
Lighting (sun, ambient, reflections, fog)	2.79	26%
Shadows (3 cascades)	1.58	14%
Post-processing	1.21	11%
Ambient occlusion	1.20	11%
Sky table	0.02	0%
GPU total	10.91	100%

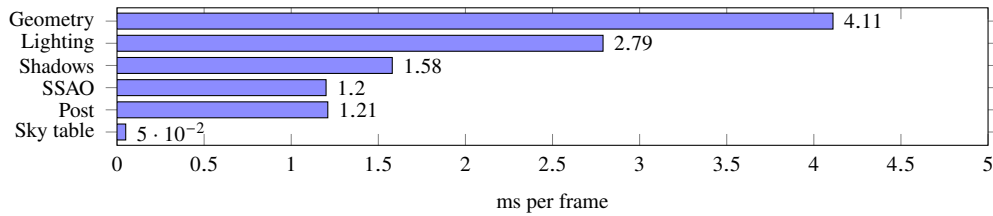


Figure 4.1: GPU time per pass (total 10.91 ms of the 16.7 ms budget).

4.5.2 Qualitative results

Figure 4.2 shows the running game: the base, a firefight with alarm, the automatic material showroom, IK-posed soldiers, tree species, drifting clouds and hands on a ladder. Panels (h) and (i) of Fig. 4.2 document a bug: the fog colour sampled the whole sky, so a horizon star painted a vertical line on every surface in its column (left of centre); the fog now uses the smooth atmosphere only.

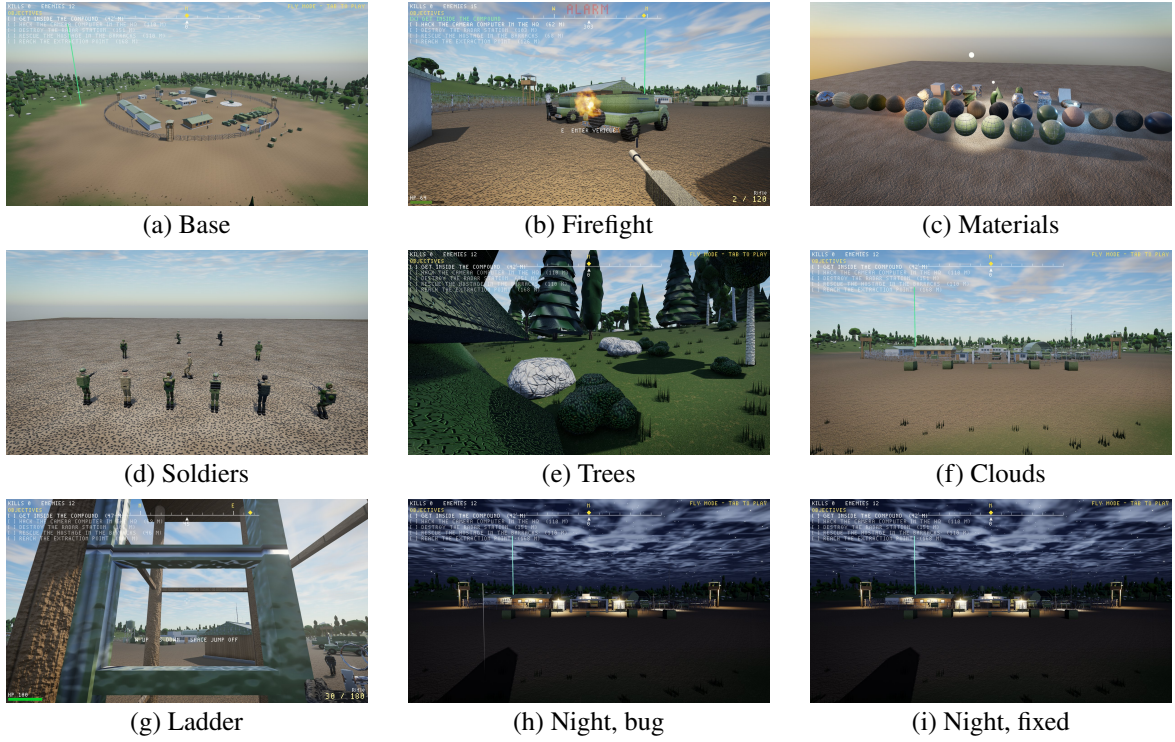


Figure 4.2: The engine and game in action; (h) and (i) show the streak bug and its fix.

4.5.3 Analysis of the results

Geometry costs most (38%), so instancing and per-chunk culling matter; the GPU uses two thirds of the budget. The streak was found by bisecting: removing draw items, lights and effects until it vanished.

4.6 Objective Achieved

(1) Six layers, downward-only. (2) No model, image or sound file. (3) Deferred PBR with shadows, SSAO, reflections, sky, bloom, ACES, FXAA. (4) Two playable missions. (5) 281 tests, 327 000 GL checks, 10.9 ms GPU time.

4.7 Conclusion

A procedural engine meets a 60 fps budget and passes exact checks.

CHAPTER V

Societal, Health, Environment, Safety, Ethical, Legal and Cultural Issues

5.1 Intellectual Property Considerations

All code, shapes, textures and sounds in Sentinel are original: because every asset is generated from formulas and seeds, there are no third-party models, images or recordings to license. The libraries used (Odin, GLFW, miniaudio) are open source under permissive licences, and the papers and notes listed in the references are cited wherever their ideas are used. The mission is inspired by, but not copied from, *Project I.G.I.*: no names, maps or art are reused.

5.2 Ethical Considerations

The game is a work of fiction about an invented compound. Its figures are abstract, low-detail shapes, and it contains no real places, groups or people. It is intended for adult players. No personal data is collected, no network connection is made, and saved progress is a small local text file that the player can read or delete.

5.3 Safety Considerations

Muzzle flashes, explosions and the alarm can bother light-sensitive players, and loud sound can hurt hearing. For this reason brightness changes are smoothed by an eye-adaptation curve, flashes are short, and all sounds are mixed and limited before reaching the speakers. Players are advised to take regular breaks, and the pause menu is always available with one key. The game also never forces fast reaction: enemies give an audible warning before an alarm, and crouching makes the player quieter and harder to see.

5.4 Legal Considerations

Open-source licence terms are respected, and no proprietary code, trademark, logo or map is included. Violence is stylised, and the project makes no claim to real military accuracy. The report credits all cited work, and the course rules on originality are followed: the code was written for this course, and the earlier course version is kept as a clearly marked git tag.

5.5 Impact on Societal, Health and Cultural Issues

An open, readable engine lowers the barrier to learning graphics: a student can read one short file per feature, change a number and see its effect at once. This supports the course aim of understanding rendering by building it, and the table-driven content lets non-programmers adjust shapes and materials. Shooter games can encourage long play sessions, so the design favours stealth, crouching and rescue over constant shooting; the player is rewarded for avoiding detection and for freeing the hostage. No cultural or religious symbols are used, and the briefing text is neutral and fictional. The tactical map and compass help players who find navigation hard.

5.6 Impact on the Environment and Sustainability

Because no art files are downloaded, the whole project is a few megabytes of source code, which reduces storage and transfer energy; a conventional game of similar scope would ship hundreds of megabytes of models and textures. Display synchronisation caps the frame rate at the monitor's refresh rate and avoids needless GPU work, and the GPU finishes each frame in about two thirds of its time budget, so the laptop is not driven at full load.

CHAPTER VI

Addressing Complex Engineering Problems and Activities

6.1 Complex Engineering Problems

Table 6.1 maps the project to the usual attributes of a complex engineering problem. The requirements pull against each other: realism needs many passes, the budget allows 16.7 ms, “no files” removes easy detail, and the platform removes compute shaders.

Table 6.1: Complex engineering problem attributes

Attribute	In this project
Depth of knowledge	Radiometry, noise theory, numerical filters, GPU pipelines, collision, state machines, sound synthesis.
Conflicting requirements	Realism versus a 16.7 ms frame; no files versus rich surfaces; OpenGL 4.1 without compute shaders.
Depth of analysis	Bisecting the fog star streak; proving that textures tile exactly; tuning physics curves.
Interdependence	Renderer, world, game and audio share data layouts and timing.
Novelty	A complete game from code and seeds only, written to be read.

6.2 Complex Engineering Activities

The project used modern resources (compiler, GPU timers, test runners, version control) and many interacting components: a fog change affected the sky, clouds, stars and night lights. Quality was verified by measurement rather than by eye alone (Fig. 6.1), with unit tests, GL checks and scripted playthroughs run before each commit.

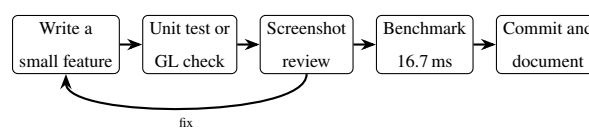


Figure 6.1: The verification loop used for every feature.

CHAPTER VII

Conclusions

7.1 Summary

Sentinel shows that a complete 3D game, from surfaces and shapes to weather and sound, can be generated entirely by code under an OpenGL 4.1 limit: a six-layer engine, a deferred physically based renderer, noise-based recipes, table-driven content, a mission game and a large test suite fit in about 21 000 lines and run within a 60 fps budget (10.9 ms of GPU time). Most of what looks like “art” is a few simple ideas applied consistently and tested.

7.2 Limitations

Models are blocky (primitives plus noise), and characters are simple. There is no true global illumination or ray tracing; indirect light and reflections are screen-space approximations. Foliage is lumpy volume, the beacon is a solid column, soldiers cannot pick up weapons, and tuning used scripts and screenshots.

7.3 Recommendations and Future Works

The engine can be extended with leaf cards with cut-out transparency for more realistic trees, screen-space global illumination, point-light shadows, anisotropic highlights for brushed metal, networked multiplayer and a larger user study of difficulty and comfort. A port to a modern graphics API with compute shaders would allow noise, terrain and even animation to be generated on the GPU, and would allow true ray-traced reflections.

APPENDIX A: EQUATION GUIDE

Each subsection explains one equation of Chapter III (bracketed numbers are the report's equation numbers): what it does, what each symbol means, why it has that shape, a worked numeric example (checked by script), the exact place in the code, and a short spoken summary. All formulas were checked against the source files.

G1. Noise deformer (3.1): turning a sphere into a rock

$$\mathbf{p}' = \mathbf{p} + A \nu(f \mathbf{p}) \hat{n}, \quad \|\mathbf{p}' - \mathbf{p}\| \leq A.$$

Meaning. Every vertex $\mathbf{p}$ of a mesh is pushed along its surface direction $\hat{n}$ by a smooth random amount. **Symbols:** A amplitude in metres (largest possible push); $\nu \in [-1, 1]$ a 3-D fractal noise value; f frequency, in cycles per metre (higher f gives smaller bumps, because the noise is sampled at $f\mathbf{p}$); $\hat{n}$ unit normal of the original surface. **Why the bound holds:** $|\nu| \leq 1$ and $\|\hat{n}\| = 1$, so the push is at most A ; this is what makes object size ranges testable. Neighbouring vertices get nearly equal ν (noise is smooth), so the surface stays connected and lumpy, not spiky.

Worked example (numbers checked by script)

$A = 0.3$ m and $\nu = 0.5$ at one vertex: the vertex moves $0.3 \times 0.5 = 0.15$ m outward. At $\nu = -1$ it moves 0.3 m inward.

Code: Engine/Procedural/Deform.odin, case Noise_Displace.

How to say it in 20 seconds

“A noise deformer moves each vertex along its normal by amplitude times smooth noise, so shapes get natural lumps while the maximum change is guaranteed to be the amplitude.”

G2. Two-bone inverse kinematics (3.2): placing a foot

$$\cos \alpha = \frac{a^2 + c^2 - b^2}{2ac}$$

Meaning. A leg has an upper bone (length a , thigh) and a lower bone (length b , shin). Given where the foot must be, we need the hip angle. The hip, knee and foot form a triangle with sides a , b and $c = |\text{target} - \text{hip}|$; the *law of cosines* gives any angle from the three sides. **Symbols:** α angle at the hip between the hip-to-target line and the thigh. The knee position then lies at distance $a \cos \alpha$ along the target line and $a \sin \alpha$ to the side. The interior knee angle θ uses the same law: $\cos \theta = (a^2 + b^2 - c^2)/2ab$. **Safety:** the code clamps c and the cosine to $[-1, 1]$, so an unreachable target gives a straight leg instead of a numerical error.

Worked example (numbers checked by script)

$a = b = 0.45$ m, foot target $c = 0.8$ m away: $\cos \alpha = 0.64/0.72 = 0.889$, so $\alpha = 27.3^\circ$ and the knee angle is 125.5° (bent but not folded). At $c = 0.9 = a + b$ the leg is straight ($\alpha = 0$).

Code: Game/Characters/Ik.odin (law of cosines in the header comment and body).

How to say it in 20 seconds

“Hip, knee and foot make a triangle whose sides I know, so the law of cosines gives the joint angle; no animation files are needed.”

G3. Fractal noise (3.3) and why textures tile

$$F(\mathbf{x}) = \frac{\sum_{i=0}^{n-1} 2^{-i} \nu(2^i \mathbf{x})}{\sum_{i=0}^{n-1} 2^{-i}}$$

Base noise ν . At each grid corner a random unit direction (one of 16, picked by a hash of the corner index) is stored. For a point inside a cell, ν takes the dot product of each corner direction with the vector from that corner to the point, then blends the four values with the smooth curve $s(t) = 6t^5 - 15t^4 + 10t^3$ (it has zero slope and zero curvature at 0 and 1, so no grid lines show), and multiplies by $\sqrt{2}$ so the result spans $[-1, 1]$. **Fractal sum.** Octave i has twice the frequency ($2^i \mathbf{x}$) and half the strength (2^{-i}): big hills, then boulders, then pebbles. The denominator is the total strength, so F stays in $[-1, 1]$. **Tiling.** The corner direction is chosen from the *cell index modulo the period P* (and P doubles each octave). Then $F(\mathbf{x} + P\mathbf{e}) = F(\mathbf{x})$ for whole-number shifts $\mathbf{e}$: the right edge of a texture equals its left edge, so repeated tiles have no seam. The test suite checks this exactly (shift $(1, -2)$ tiles; difference must be 0).

Worked example (numbers checked by script)

$n = 4$ octaves: weights $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$ sum to 1.875. A value $\nu = +1$ in every octave gives $F = 1$; typical values mix and give about ± 0.4 . $s(0.5) = 0.5$ exactly, and $s(0.25) = 0.104$, so the blend is gentle near cell edges.

Code: Shaders/Include/Noise.glsl (gradient_noise, fbm).

How to say it in 20 seconds

“Fractal noise adds smooth random layers, each twice as detailed and half as strong; by wrapping the grid index it repeats exactly, so textures tile.”

Variants used: *domain warping* $F(\mathbf{x} + s[F_1(\mathbf{x}), F_2(\mathbf{x})])$ evaluates F at a point shifted by two other noises, which bends straight lattice patterns into swirls (still periodic, since every term is). *Ridged* noise uses $(1 - |\nu|)^2$: $|\nu|$ folds the noise at zero so zero crossings become sharp ridges (cracks, bark). *Cellular* (Worley) noise uses the distance to the nearest random point (leaves, pebbles).

G4. Normal map from a height map (3.4)

$$h_x \approx \frac{3(h_{NE} + h_{SE} - h_{NW} - h_{SW}) + 10(h_E - h_W)}{32 \Delta N}, \quad \hat{n} = \frac{(-k h_x, -k h_y, 1)}{\|(-k h_x, -k h_y, 1)\|}$$

Meaning. A recipe provides a height h per texel. Lighting only needs the surface *direction*, which tilts away from the slope: a surface rising to the right ($h_x > 0$) tilts its normal to the left ($-h_x$). **Scharr filter:** the slope is a weighted difference of the right column (3, 10, 3) minus the left column of the 3×3 neighbourhood. Weights sum to 16 over a distance of two texels, hence the divisor 32Δ ; it treats diagonal and axis-aligned slopes equally, which the plain two-point difference does not. **Symbols:** Δ one texel (here $1/1024$ of the tile); N cells per tile (so a bumpy recipe with more cells is not made steeper); k the per-material bump strength; the vector is normalised to unit length. The last component 1 means a flat area gives $\hat{n} = (0, 0, 1)$.

Worked example (numbers checked by script)

A perfect ramp $h = 0.7x$ (with $\Delta = 1$): the numerator is $3 \cdot 2.8 + 10 \cdot 1.4 = 22.4$, divided by 32 gives 0.7, the true slope (verified by script). With $k = 1$: $\hat{n} = (-0.7, 0, 1)/1.22 = (-0.57, 0, 0.82)$, tilted 35° .

Cavity darkening. Where h is below the mean of its four neighbours (a negative Laplacian), the albedo is multiplied by $1 - 0.5c$ and the ambient occlusion by $1 - c$, where $c \in [0, 0.5]$ grows with the depth of the dent. **Triplanar weights:** $w_i = |n_i|^4 / \sum_j |n_j|^4$ (the code uses exponent 4): a surface facing $+y$ mostly uses the top-down projection. For $\mathbf{n} = (0.6, 0.8, 0)$ the weights are 0.24, 0.76, 0. **Code:** Shaders/Include/BakeMain.glsl, Shaders/Include/Triplanar.glsl.

How to say it in 20 seconds

“The height map’s slope tells how the surface tilts; a Scharr filter measures it fairly in every direction, and the normal is minus the slope with a one up, normalised.”

G5. Cook–Torrance reflection (3.5)

$$f = \frac{(1 - F_l)(1 - F_v) \mathbf{c}_d}{\pi} + D V F, \quad D = \frac{\alpha^2}{\pi((\mathbf{n} \cdot \mathbf{h})^2(\alpha^2 - 1) + 1)^2}, \quad F = F_0 + (1 - F_0)(1 - \mathbf{v} \cdot \mathbf{h})^5$$

$$V = \frac{0.5}{(\mathbf{n} \cdot \mathbf{l})\sqrt{(\mathbf{n} \cdot \mathbf{v})^2(1 - \alpha^2) + \alpha^2} + (\mathbf{n} \cdot \mathbf{v})\sqrt{(\mathbf{n} \cdot \mathbf{l})^2(1 - \alpha^2) + \alpha^2}}$$

Meaning. f is the fraction of light arriving from direction $\mathbf{l}$ that leaves toward the viewer $\mathbf{v}$; the lighting pass multiplies it by the light’s brightness and $\mathbf{n} \cdot \mathbf{l}$. The surface is imagined as many tiny mirrors (microfacets). **Vectors:** $\mathbf{n}$ surface normal, $\mathbf{l}$ to the light, $\mathbf{v}$ to the eye, $\mathbf{h}$ halfway between $\mathbf{l}$ and $\mathbf{v}$ (only facets facing $\mathbf{h}$ reflect light from $\mathbf{l}$ to $\mathbf{v}$). **Specular part DVF .** D (GGX): the share of facets facing $\mathbf{h}$; rough surfaces have small peak, wide lobe. $\alpha = \text{roughness}^2$. $V = G/(4 \mathbf{n} \cdot \mathbf{l} \mathbf{n} \cdot \mathbf{v})$: the geometry term G (facets blocking each other) with the denominator folded in (height-correlated Smith form). F (Schlick): every surface reflects more at glancing angles; F_0 is reflectance looking straight on (0.04 for plastic, stone, paint; the albedo colour for metals). **Diffuse part.** $\mathbf{c}_d/\pi$ is the Lambert term (π makes total reflected light equal the colour). It is scaled by $(1 - F_l)(1 - F_v)$ because light that mirror-reflected at the surface cannot also scatter inside; this keeps energy ≤ 1 . Metals have $\mathbf{c}_d = 0$.

Worked example (numbers checked by script)

Roughness 0.5 gives $\alpha = 0.25$. Peak of D at $\mathbf{n} \cdot \mathbf{h} = 1$: $1/(\pi \cdot 0.0625) = 5.09$. Fresnel with $F_0 = 0.04$: looking at 60° gives $F = 0.07$; at 85° (grazing) $F = 0.65$, why wet roads shine at low angles.

Code: Shaders/Include/Brdf.gls1, microfacet_cook_torrance.

How to say it in 20 seconds

“Light reflects as a diffuse part plus a specular part from microscopic mirrors: GGX says how many face the eye, Smith says how many hide each other, Schlick says how reflective at this angle.”

G6. Fog by distance (haze)

$$C = T C_{\text{surface}} + (1 - T) C_{\text{haze}}, \quad T = e^{-kd}, \quad k = 3.5 \times 10^{-4} \text{ m}^{-1}$$

Meaning. Air absorbs and scatters light. T is the fraction of a surface’s light that survives d metres of air (Beer–Lambert law: equal thickness removes an equal *fraction*, so the decay is exponential); the missing part is replaced by the sky-coloured haze, a linear mix. **Symbols:** d distance to the camera; k density.

Worked example (numbers checked by script)

$d = 100 \text{ m}$: $T = 0.966$ (almost clear). $d = 1 \text{ km}$: $T = 0.705$. $d = 3 \text{ km}$: $T = 0.35$, so far hills are mostly sky colour, giving depth. The haze colour uses only the smooth atmosphere (never stars or the moon), which fixed the night streak bug.

Code: Shaders/DeferredBase.gls1 (FOG_DENSITY_PER_METER).

How to say it in 20 seconds

“Each metre of air removes the same percentage of light, so transmittance is an exponential, and what is lost is replaced by sky colour.”

G7. ACES tone mapping (3.6)

$$C_{\text{out}} = \text{clamp}_{[0,1]} \left(\frac{x(2.51x + 0.03)}{x(2.43x + 0.59) + 0.14} \right), \quad x = \text{exposure} \times \text{radiance}$$

Meaning. Scene light (HDR) can be 0.01 in shade or 40 at the sun, but a display shows 0 to 1. This curve (Narkowicz’s fit of the ACES film curve) compresses highlights smoothly (shoulder), lifts dark detail gently (toe) and has an S-shaped contrast like film. **Symbols:** x radiance scaled by exposure (1 by day, rising to 6 at night, like the eye adapting); applied to each colour channel. The numbers are fitted constants.

Worked example (numbers checked by script)

$x = 0.18$ (mid-grey) $\rightarrow 0.267$; $x = 1 \rightarrow 0.804$; $x = 4 \rightarrow 0.973$. Quadrupling brightness from 1 to 4 changes the output by only 0.17: highlights do not clip to white abruptly.

After the curve the image is graded, converted to sRGB for the display and dithered by about one 8-bit step to avoid banding. **Code:** Shaders/Include/Tonemap.glsl, Shaders/PostTonemap.glsl.

How to say it in 20 seconds

“A rational curve maps unlimited brightness into zero-to-one with a soft shoulder, the way film does.”

G8. Sun height (inline in Section 3.2.5)

$$\sin(\text{elev}) = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos h, \quad h = 15^\circ (\text{hours} - 12)$$

Meaning. Standard spherical astronomy. $\varphi = 35^\circ$ latitude, $\delta = 15^\circ$ the sun’s declination (a late-spring day), h the hour angle (the Earth turns 15° per hour; $h = 0$ at noon). The full direction vector uses the same terms with east and south components. Positive result: sun above the horizon.

Worked example (numbers checked by script)

Noon: $\sin = 0.1485 + 0.7913 = 0.9398$, elevation 70° . At 6:00 and 18:00: 8.5° (low, orange sun). Midnight: -40° , so the moon (opposite the sun) lights the scene.

Code: Game/Gameplay/TimeOfDay.odin.

G9. Cloud plane and wind (3.7)

$$\mathbf{P}_{xz} = \frac{H}{d_y} (d_x, d_z) + \mathbf{w}t$$

Meaning. Clouds are drawn as a flat noise layer at height H . For a view direction $\mathbf{d} = (d_x, d_y, d_z)$, a ray from the ground reaches height H after travelling H/d_y times the direction (similar triangles); that gives the horizontal position $(d_x, d_z)H/d_y$ where the noise is looked up. Adding $\mathbf{w}t$ ($\mathbf{w}$ wind velocity in m/s, t time) slides the pattern, so clouds drift. A faster detail layer ($\times 1.35$) and a cirrus layer at 7 km ($\times 1.8$ speed) make shapes change and give depth. Rays near the horizon ($d_y < 0.04$) are skipped, because H/d_y would blow up.

Worked example (numbers checked by script)

$\mathbf{d} = (0.8, 0.5, 0.33)$ and $H = 1500$: $H/d_y = 3000$, point (2400, 995) m. With $\mathbf{w} = (14, 5)$ m/s the pattern moved (840, 300) m after one minute.

Code: Shaders/Include/Sky.glsl (add_clouds, cloud_density); wind in Game/Sandbox/Sandbox.odin.

G10. Tree sway (cantilever wind)

$$\Delta_x = 2s(0.04 + 0.1g)\left(\frac{y}{9}\right)^2, \quad \Delta_z = 0.35\Delta_x, \quad g = \frac{1}{2} + \frac{1}{2}\sin(1.3t + 0.05x + 0.03z)$$

Meaning. The vertex shader shifts each tree vertex sideways. A real tree is like a beam fixed at the root: it barely moves at the bottom and most at the top, so the shift grows with the *square* of height y (clamped to 9 m). $g \in [0, 1]$ is a gust that travels across the land (position terms $0.05x + 0.03z$ delay it from tree to tree, so trees do not sway in lockstep). s is the per-mesh sway (0.6 for trunks, 1 for foliage; foliage above 1 also adds a small fast flutter). The factor 2 and offsets 0.04/0.1 are tuned amplitudes: calm sway 0.08 m, full gust 0.28 m at the top.

Worked example (numbers checked by script)

Foliage ($s = 1$), full gust $g = 1$: at $y = 9$ m, $\Delta_x = 2(0.14)(1) = 0.28$ m; at half height 4.5 m it is 0.07 m (a quarter, because of the square).

Code: Shaders/Geometry.glsl, wind_offset.

G11. Vehicle steering (bicycle model) and engine power

$$\dot{\psi} = \frac{v \tan \delta}{L}, \quad \dot{v} = a_0 \left(1 - 0.6 \frac{|v|}{v_{\max}}\right)$$

Steering. Treat the car as a bicycle with wheel base L (m), front wheel angle δ , speed v . The car turns on a circle of radius $R = L/\tan \delta$, so the yaw rate is $v/R = v \tan \delta/L$ (rad/s). Faster means turning faster; reversing flips the sign, as in a real car. (Tracked vehicles instead turn at a fixed rate and can pivot.) **Power fade.** Engine thrust is power divided by speed, so acceleration is high from rest and falls as speed rises. a_0 is the maximum acceleration, $v_{\max}$ the limit; at top speed acceleration is still $0.4a_0$ but the speed is clamped to the limit.

Worked example (numbers checked by script)

Jeep ($L = 3$, $\delta_{\max} = 0.55$ rad, $v_{\max} = 18$, $a_0 = 2.6$): at 10 m/s full lock gives $\dot{\psi} = 10 \tan 0.55/3 = 2.04$ rad/s. Solving the speed equation, $v(t) = 30(1 - e^{-0.0867t})$: 2.5 m/s after 1 s, and 18 m/s after 10.6 s (it was 24 m/s with 7 m/s² before the realism change).

Code: Engine/World/GroundVehicle.odin (yaw_rate_for, next_speed).

Short answers to likely questions

- **Why not use image textures?** The project rule forbids files; noise gives infinite variety from a seed and tiles exactly.
- **Why a quintic fade and not linear blending?** Linear blending shows grid lines (slope jumps at cell edges); $6t^5 - 15t^4 + 10t^3$ has zero first and second derivative there.
- **Why $\sqrt{2}$ in the noise?** Unit gradients bound the raw value by $\sqrt{2}/2 \approx 0.707$; multiplying by $\sqrt{2}$ makes the range $[-1, 1]$.
- **Why exponential fog?** Beer–Lambert: each metre removes the same fraction of light.
- **Why is the GPU time 10.9 ms but the frame 16.7 ms?** The display waits for the 60 Hz refresh; the extra 5.8 ms is idle headroom.
- **What did you correct after checking?** IK used the root angle (law of cosines on a, c, b), Cook–Torrance has a visibility term V and a diffuse part, the normal-map slope is divided by cells per tile, and the wind has exact constants.

REFERENCES

- [1] K. Perlin, “Improving noise,” *ACM Trans. Graph. (SIGGRAPH)*, vol. 21, no. 3, pp. 681–682, 2002.
- [2] S. Worley, “A cellular texture basis function,” in *Proc. SIGGRAPH*, 1996, pp. 291–294.
- [3] R. L. Cook and K. E. Torrance, “A reflectance model for computer graphics,” *ACM Trans. Graph.*, vol. 1, no. 1, pp. 7–24, 1982.
- [4] B. Walter, S. R. Marschner, H. Li and K. E. Torrance, “Microfacet models for refraction through rough surfaces,” in *Proc. Eurographics Symp. Rendering*, 2007, pp. 195–206.
- [5] B. Karis, “Real shading in Unreal Engine 4,” SIGGRAPH Course: Physically Based Shading in Theory and Practice, 2013.
- [6] S. Hillaire, “A scalable and production ready sky and atmosphere rendering technique,” *Computer Graphics Forum*, vol. 39, no. 4, pp. 13–22, 2020.
- [7] R. Dimitrov, “Cascaded shadow maps,” NVIDIA White Paper, 2007.
- [8] M. Oren and S. K. Nayar, “Generalization of Lambert’s reflectance model,” in *Proc. SIGGRAPH*, 1994, pp. 239–246.
- [9] K. Narkowicz, “ACES filmic tone mapping curve,” blog post, 2016.
- [10] J. Jimenez, “Next generation post processing in Call of Duty: Advanced Warfare,” SIGGRAPH Course: Advances in Real-Time Rendering, 2014.
- [11] T. Lottes, “FXAA,” NVIDIA White Paper, 2009.
- [12] M. Segal and K. Akeley, *The OpenGL Graphics System: A Specification, Version 4.1*, Khronos Group, 2010.
- [13] I. Quilez, “Domain warping,” online article, iquilezles.org.
- [14] S. Hargreaves and M. Harris, “Deferred shading,” NVIDIA presentation, Game Developers Conference, 2004.
- [15] H. Scharr, “Optimal operators in digital image processing,” Ph.D. dissertation, Univ. Heidelberg, 2000.
- [16] T. Akenine-Möller, E. Haines and N. Hoffman, *Real-Time Rendering*, 4th ed., CRC Press, 2018.